

Radical Axes

Problem 1. We are given a cyclic quadrilateral $ABCD$ with a point E on the diagonal AC such that $AD = AE$ and $CB = CE$. Let M be the centre of the circumcircle k of the triangle BDE . The circle k intersects the line AC in the points E and F . Prove that the lines FM , AD and BC meet at one point.

Problem 2. Given a triangle ABC , let P and Q be points on segments AB and AC , respectively, such that $AP = AQ$. Let S and R be distinct points on segment BC such that S lies between B and R , $\angle BPS = \angle PRS$, and $\angle CQR = \angle QSR$. Prove that P, Q, R, S are concyclic.

Problem 3. Let P be a point inside an acute triangle ABC . Let the lines BP and CP intersect the sides AC and AB at D and E , respectively. Let the circles with diameters BD and CE intersect at points S and T . Prove that if the points A, S , and T are collinear, then P lies on a median of $\triangle ABC$.

Problem 4. Let $\triangle ABC$ be a scalene triangle with incentre I and circumcircle ω . Lines AI, BI, CI intersect ω for the second time at points D, E, F , respectively. The parallel lines from I to the sides BC, AC, AB intersect EF, DF, DE at points K, L, M , respectively. Prove that the points K, L, M are collinear.

Problem 5. Let ABC be a triangle with $AB < AC$. Let ω be a circle passing through B, C and assume that A is inside ω . Suppose X, Y lie on ω such that $\angle BXA = \angle AYC$. Suppose also that X and C lie on opposite sides of the line AB and that Y and B lie on opposite sides of the line AC . Show that, as X, Y vary on ω , the line XY passes through a fixed point.

Problem 6. In $\triangle ABC$, let D, E, F be points on sides BC, CA, AB respectively such that $FE \parallel BC$. The line DE meets the circumcircle of $\triangle ACD$ again at P . The line DF meets the circumcircle of $\triangle ABD$ again at Q . Let X be the symmetric point of D about the midpoint of BC . Prove that P, Q, D, X are concyclic.

Problem 7. Let ABC be an acute triangle with orthocentre H and circumcircle Γ . A line through H intersects segments AB and AC at E and F , respectively. Let K be the circumcentre of $\triangle AEF$, and suppose line AK intersects Γ again at a point D . Prove that line HK and the line through D perpendicular to BC meet on Γ .

Problem 8. In triangle ABC , with $AB < AC$, I is the incentre, E is the intersection of A -excircle and BC . Point F lies on the external angle bisector of $\angle BAC$ such that E and F lie on the same side of the line AI and $\angle AIF = \angle AEB$. Point Q lies on BC such that $\angle AIQ = 90^\circ$. Circle ω_b is tangent to FQ and AB at B , circle ω_c is tangent to FQ and AC at C and both circles pass through the inside of triangle ABC . If M is the midpoint of the arc BC of the circumcircle of triangle ABC which does not contain A , prove that M lies on the radical axis of ω_b and ω_c .